

Question 1: IT Support Cost Optimization

Data Analytics Internship Assignment

1. Problem Statement

The company handles around 2 million IT support tickets per year. The goals are to reduce total support cost by 20%, improve customer satisfaction above 4.2, reduce unnecessary escalations and optimize engineer allocation.

2. Objectives

- Identify support cost drivers.
- Analyze resolution time and escalations.
- Check customer satisfaction.
- Compare cost by issue category and priority.
- Give practical cost-reduction recommendations.

3. Data Used and Cleaning

A sample dataset of 100,000 tickets was created because the actual company dataset was not provided. It contains issue category, priority, resolution time, engineer hourly cost, client type, escalation count and customer satisfaction. Missing numeric values were filled using median imputation. Duplicate records were checked and none were found.

4. Overall Cost

Total sample support cost = ₹295,858,500. Average cost per ticket = ₹2,958.59.

5. Cost by Issue Category

Category	Total Cost (₹)	Average Cost (₹)
Infrastructure	59,901,146	2,967.31
Software	59,204,335	2,947.93
Security	59,120,005	2,957.91
Network	58,964,854	2,945.59
Hardware	58,690,060	2,974.36

Infrastructure has the highest total cost, while Hardware has the highest average cost per ticket.

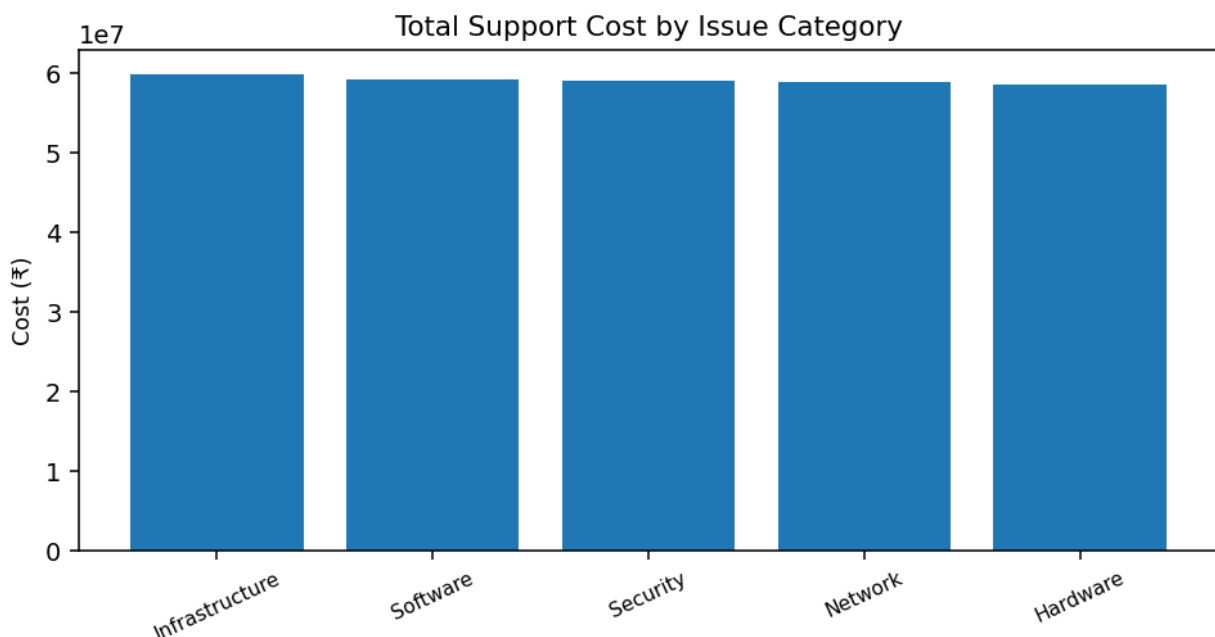


Figure 1: Total support cost by issue category.

6. Cost by Priority

Priority	Total Cost (₹)	Average Cost (₹)
Medium	117,966,003	2,945.03
Low	88,521,814	2,965.65
High	74,678,956	2,969.81
Critical	14,691,727	2,968.63

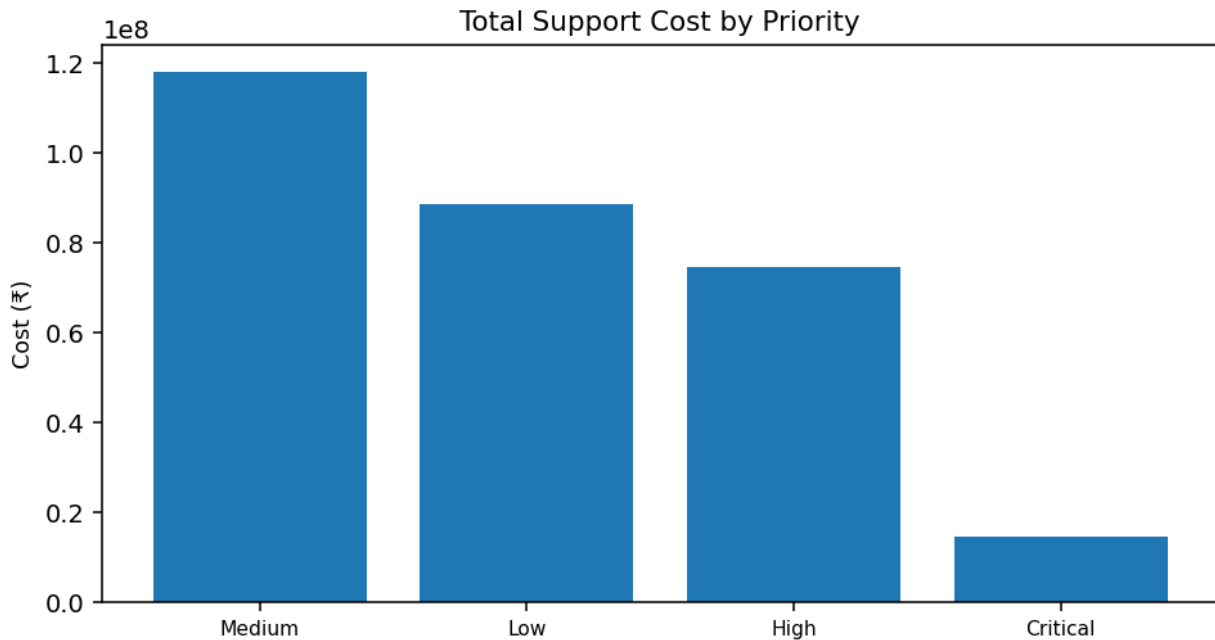


Figure 2: Total support cost by priority.

Medium-priority tickets have the highest total cost mainly because of their higher volume.

7. Resolution Time

Hardware has the highest average resolution time at about 4.95 hours. Network tickets have the highest maximum resolution time at about 33.38 hours.

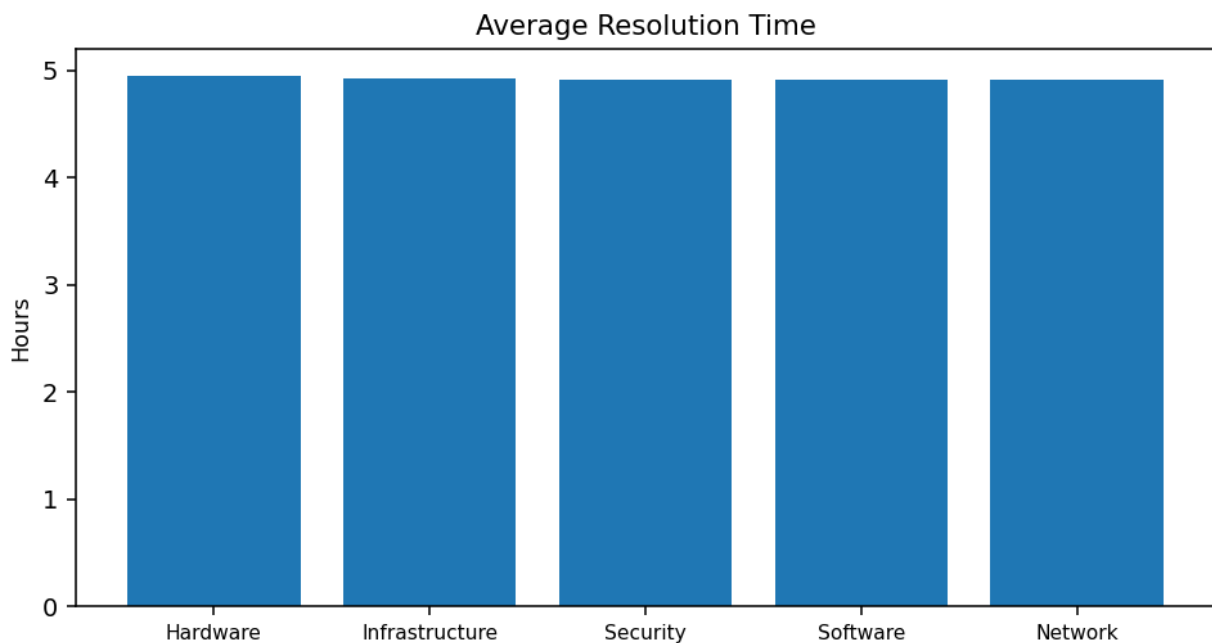


Figure 3: Average resolution time.

8. Customer Satisfaction

Average satisfaction is very similar across issue categories. Security is highest at about 4.004 and Network is lowest at about 3.994.

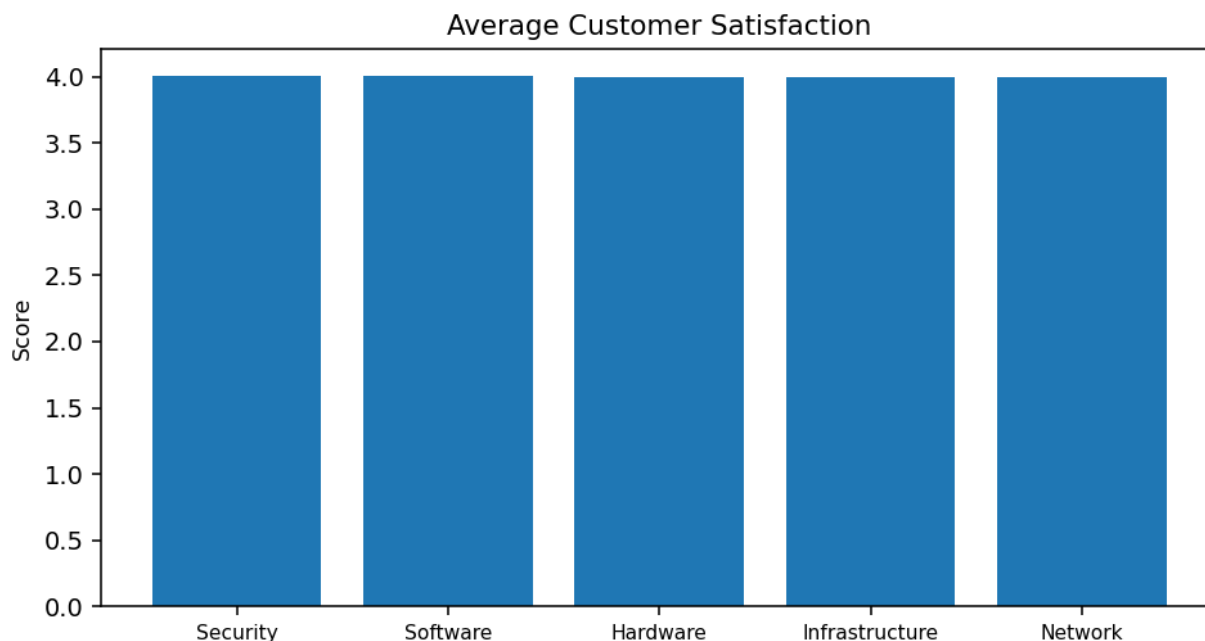


Figure 4: Average customer satisfaction.

9. Escalation Analysis

Escalation counts were analyzed by issue category to identify areas where unnecessary escalations may be reduced. This can help in improving first-contact resolution and engineer allocation.

10. Main Findings

- Infrastructure is the largest total cost area.
- Medium priority contributes the highest overall cost.
- Network has the longest maximum resolution time.
- Satisfaction is relatively consistent across categories.
- Reducing resolution time and improving engineer allocation can help reduce cost.

11. Recommendations

- Focus cost-reduction efforts on Infrastructure.
- Allocate engineers according to ticket volume and complexity.
- Investigate long-running Network issues.
- Use automation/self-service for repetitive requests.
- Monitor cost, resolution time, escalations and satisfaction regularly.

12. Conclusion

The analysis shows that support cost can be reduced by focusing on high-cost areas, high-volume priorities, long-running tickets and repetitive support work. These actions can help the company move toward the 20% cost-reduction target.

Note: Sample/synthetic data was used for this assignment.